

[illegible]

name	local test	public test	desc	architecture	result
v24b pure node2d	0.5395	mistake		v24b Per-class accuracy: Confounder : 0.5396 (n=682) Collider : 0.4462 (n=446) Mediator : 0.4513 (n=483) Cause of X : 0.6236 (n=983) Cause of Y : 0.6867 (n=2145) Consequence of X : 0.4976 (n=1571) Consequence of Y : 0.5806 (n=682) Independent : 0.5000 (n=4494)	
v25 stacked curves	0.7764	conv2d on stacked curves		v24b Balanced Accuracy: 0.5395 v25 Per-class accuracy: Confounder : 0.7258 (n=682) Collider : 0.7534 (n=446) Mediator : 0.6390 (n=483) Cause of X : 0.7904 (n=983) Cause of Y : 0.8448 (n=2145) Consequence of X : 0.7944 (n=1571) Consequence of Y : 0.7962 (n=682) Independent : 0.8870 (n=4494)	
v25b scatter density	0.7899			v25 Balanced Accuracy: 0.7764 v25b Per-class accuracy: Confounder : 0.7375 (n=682) Collider : 0.7668 (n=446) Mediator : 0.6770 (n=483) Cause of X : 0.8159 (n=983) Cause of Y : 0.8415 (n=2145) Consequence of X : 0.7931 (n=1571) Consequence of Y : 0.8094 (n=682) Independent : 0.8781 (n=4494)	
v25a	0.7956	conv2d on the smoothed heatmap of v->X and v->Y and Y->v and X->v		v25b Balanced Accuracy: 0.7899 v25a Per-class accuracy: Confounder : 0.7566 (n=682) Collider : 0.7691 (n=446) Mediator : 0.6915 (n=483) Cause of X : 0.8210 (n=983) Cause of Y : 0.8401 (n=2145) Consequence of X : 0.7868 (n=1571) Consequence of Y : 0.8182 (n=682) Independent : 0.8814 (n=4494)	
v26b 8 channel	0.7917	conv2d pairwise, the edge head actually has a meaning now		v25a Balanced Accuracy: 0.7956 v26b Per-class accuracy: Confounder : 0.7595 (n=682) Collider : 0.7601 (n=446) Mediator : 0.6687 (n=483) Cause of X : 0.8169 (n=983) Cause of Y : 0.8522 (n=2145) Consequence of X : 0.7899 (n=1571) Consequence of Y : 0.8123 (n=682) Independent : 0.8736 (n=4494)	
v26c pure node2d	0.5151	node2d only, no edge module, this is missing the whole graph context		v26b Balanced Accuracy: 0.7917 v26c Per-class accuracy: Confounder : 0.5000 (n=682) Collider : 0.3655 (n=446) Mediator : 0.4099 (n=483) Cause of X : 0.5070 (n=983) Cause of Y : 0.6709 (n=2145) Consequence of X : 0.5169 (n=1571) Consequence of Y : 0.5396 (n=682) Independent : 0.5307 (n=4494)	
v27 edge2d	0.6522	we make the edge feature 2d instead of node feature extractor		v26c Balanced Accuracy: 0.5151 v27 Per-class accuracy: Confounder : 0.6290 (n=682) Collider : 0.5336 (n=446) Mediator : 0.5673 (n=483) Cause of X : 0.7121 (n=983) Cause of Y : 0.7315 (n=2145) Consequence of X : 0.6022 (n=1571) Consequence of Y : 0.7141 (n=682) Independent : 0.7279 (n=4494)	
v28 node attn	0.125	only node head, no edge head		v27 Balanced Accuracy: 0.6522 v28 Per-class accuracy: Confounder : 0.0000 (n=682) Collider : 0.0000 (n=446) Mediator : 0.0000 (n=483) Cause of X : 0.0000 (n=983) Cause of Y : 0.0000 (n=2145) Consequence of X : 0.0000 (n=1571) Consequence of Y : 0.0000 (n=682) Independent : 1.0000 (n=4494)	
v28b node deep sets	0.4639	same with above, the model missing the context		v28 Balanced Accuracy: 0.1250 v28b Per-class accuracy: Confounder : 0.4370 (n=682) Collider : 0.2377 (n=446) Mediator : 0.3313 (n=483) Cause of X : 0.5738 (n=983) Cause of Y : 0.6783 (n=2145) Consequence of X : 0.4780 (n=1571) Consequence of Y : 0.4677 (n=682) Independent : 0.5078 (n=4494)	
v26d favor edge loss	0.8037	pairwise edge image more channels + favor edge head loss, rougher sigma for attn, total_loss = edge loss + 0.3 * node loss, attn sigma = 2		v28b Balanced Accuracy: 0.4639 v26 Per-class accuracy: Confounder : 0.7801 (n=682) Collider : 0.7578 (n=446) Mediator : 0.6832 (n=483) Cause of X : 0.8271 (n=983) Cause of Y : 0.8597 (n=2145) Consequence of X : 0.8001 (n=1571) Consequence of Y : 0.8314 (n=682) Independent : 0.8901 (n=4494)	
				v26 Balanced Accuracy: 0.8037	

